

Your Code

```
1 #include <stdio.h>
2 int N;
3 int board[20][20];
4 int columnUsed[20];
5 int diagonal1Used[40];
6 int diagonal2Used[40];
7 int fixedRow, fixedCol;
8 int isSafe(int row, int col)
9 {
10     int d1 = row - col + N - 1;
11     int d2 = row + col;
12
13     if (columnUsed[col])
14         return 0;
15
16     if (diagonal1Used[d1])
17         return 0;
18
19     if (diagonal2Used[d2])
20         return 0;
21
22     return 1;
23 }
24 int solve(int row)
25 {
26     int col;
```

Input

```
4
1
2
```

Output

```
Enter board size N: Enter fixed
queen row (1-4): Enter fixed queen
column (1-4):
Valid Queen Arrangement:
0
```

```
26     int col;
27     int d1, d2;
28     if (row == N)
29         return 1;
30     if (row == fixedRow)
31         return solve(row + 1);
32
33     for (col = 0; col < N; col++)
34     {
35         if (isSafe(row, col))
36         {
37             d1 = row - col + N - 1;
38             d2 = row + col;
39             board[row][col] = 1;
40             columnUsed[col] = 1;
41             diagonal1Used[d1] = 1;
42             diagonal2Used[d2] = 1;
43             if (solve(row + 1))
44                 return 1;
45             board[row][col] = 0;
46             columnUsed[col] = 0;
47             diagonal1Used[d1] = 0;
48             diagonal2Used[d2] = 0;
49         }
50     }
51
52     return 0;
53 }
```

```
54 void printBoard()
55 {
56     int i, j;
57
58     printf("\nValid Queen Arrangement:\n\n");
59
60     for (i = 0; i < N; i++)
61     {
62         for (j = 0; j < N; j++)
63         {
64             if (board[i][j] == 1)
65                 printf(" Q ");
66             else
67                 printf(" . ");
68         }
69         printf("\n");
70     }
71 }
72
73 int main()
74 {
75     int d1, d2;
76
77     printf("Enter board size N: ");
78     scanf("%d", &N);
79
80     printf("Enter fixed queen row (1-%d): ", N);
81     scanf("%d", &fixedRow);
82
```

```

83     printf("Enter fixed queen column (1-%d): ", N);
84     scanf("%d", &fixedCol);
85     fixedRow--;
86     fixedCol--;
87     if (N <= 0 || N > 20)
88     {
89         printf("Invalid board size.\n");
90         return 0;
91     }
92
93     if (fixedRow < 0 || fixedRow >= N ||
94         fixedCol < 0 || fixedCol >= N)
95     {
96         printf("Invalid fixed queen position.\n");
97         return 0;
98     }
99     board[fixedRow][fixedCol] = 1;
100
101     columnUsed[fixedCol] = 1;
102
103     d1 = fixedRow - fixedCol + N - 1;
104     d2 = fixedRow + fixedCol;
105
106     diagonal1Used[d1] = 1;
107     diagonal2Used[d2] = 1;
108     if (solve(0))
109     {
110         printBoard();
111     }

```

```
101     columnUsed[fixedCol] = 1;
102
103     d1 = fixedRow - fixedCol + N - 1;
104     d2 = fixedRow + fixedCol;
105
106     diagonal1Used[d1] = 1;
107     diagonal2Used[d2] = 1;
108     if (solve(0))
109     {
110         printBoard();
111     }
112     else
113     {
114         printf("\nNo valid solution exists.\n");
115     }
116
117     return 0;
118 }
```

 Submitted